

**EXPERIMENT 6 - INTERFACING A PLC WITH A MICROCONTROLLER
AND A PC OVER ETHERNET/IP**

MCTA 3203 - MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

GROUP 11

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Abstract

This experiment focuses on interfacing a Programmable Logic Controller (PLC) with an Arduino microcontroller over Ethernet/IP using the OpenPLC Editor. The objective is to develop, simulate, and upload ladder diagram programs that perform basic control operations, such as LED blinking and a Start–Stop control circuit. The experiment begins by creating a simple ladder diagram using OpenPLC, assigning variables, compiling the program, and transferring it to an Arduino board. Hardware components such as push buttons, resistors, and LEDs are integrated to verify the real-world operation of the logic. The key findings demonstrate successful communication between the PLC environment and the microcontroller, enabling real-time control through correctly mapped inputs and outputs. The Start–Stop circuit operates reliably when implemented on Arduino, validating the functionality of the ladder logic. Overall, the experiment highlights the effectiveness of OpenPLC as an educational platform for industrial automation and microcontroller integration.

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Introduction

The purpose of this experiment is to explore how a PLC system can interface with a microcontroller and a PC through Ethernet/IP using the OpenPLC platform. PLCs are widely used in industrial automation to control electromechanical systems, and understanding how they communicate with microcontrollers is essential for modern mechatronics applications. This experiment uses the OpenPLC Editor to create ladder logic diagrams, simulate them, and upload the program to an Arduino microcontroller.

Background concepts include ladder diagrams, digital I/O addressing, and PLC-to-microcontroller communication. Ladder logic is a graphical programming language commonly used in PLCs, allowing users to represent control circuits such as Start–Stop systems.

The expected outcome of this experiment is that the ladder diagram programs will compile successfully, run in the simulation environment, and be executed by the Arduino hardware, resulting in correct LED blinking and proper operation of the Start–Stop control circuit.

Materials and Equipment

Software

1. Open PLC Editor
2. PC / Laptop with USB connection

Hardware

1. Arduino board (e.g., Arduino Mega)
2. LED
3. 2 Push Button switches
4. Resistors
5. Jumper wires
6. Breadboard

Experimental Setup

The experiment begins with configuring the OpenPLC Editor to create and simulate ladder logic diagrams. The Arduino board is connected to the PC through USB, and the correct COM port is identified for program uploading.

For the LED blinking task, the ladder diagram consists of a simple contact and a coil. Variables are created and associated with specific Arduino digital pins using OpenPLC’s physical addressing (e.g., %QX0.0 for digital output).

For the Start–Stop control circuit, two push buttons are wired to the Arduino as inputs, and an LED is connected as the output. The hardware is assembled on a breadboard following the diagram provided in the file (Fig. 6), ensuring correct connections for power rails, buttons, resistors, and LED polarity.

The ladder logic for the Start–Stop circuit is created based on the diagram (Fig. 5), compiled, and uploaded to the Arduino. Once uploaded, the Arduino runs the PLC-created logic, enabling real-time control through physical inputs.

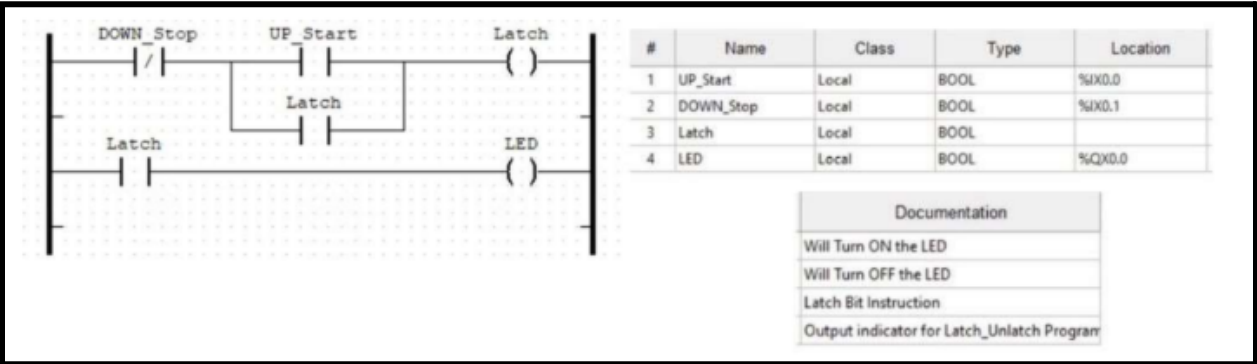


Fig 5 : Ladder diagram for the Start-Stop Circuit

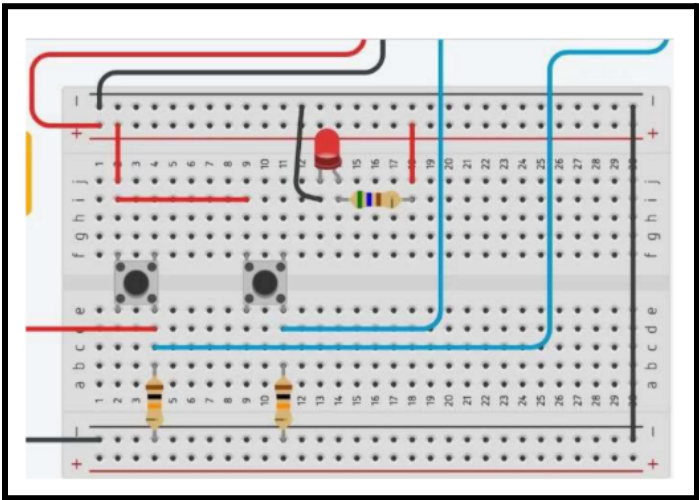


Fig 6

Methodology

Step 1: Creating Ladder Diagram in OpenPLC Editor

- Download and open the OpenPLC Editor.
- Start a new project by creating a new folder and naming it.
- In the POU creation window, name the project and select **LD (Ladder Diagram)**.
- Add new variables (Class: Local, Type: BOOL).
- Construct the ladder diagram using:
 - Left and right power rails
 - Contact (normal or negated)
 - Coil
- Compile the diagram and verify that the simulation runs correctly.

Step 2: Assigning Physical Addresses

1. Identify the correct Arduino digital output pin from the “Physical Addressing” reference.
2. Assign the location (e.g., %QX0.0) to the variable controlling the LED.
3. Connect the Arduino to the laptop and check the COM port.

Step 3: Uploading Program to Arduino

1. Click the upload icon in OpenPLC.
2. Select the correct board and COM port.
3. Transfer the ladder diagram program to the Arduino.
4. Observe the LED blinking or circuit operation.

Step 4: Start–Stop Circuit Implementation

1. Build the Start–Stop hardware circuit using two push buttons, resistors, and an LED.
2. Create the Start–Stop ladder logic (as shown in Fig. 5 of the file).
3. Compile, simulate, and upload to Arduino.
4. Test the circuit to ensure proper response to Start and Stop inputs.

Data Collection

Instruments and Hardware Used

The following hardware and software were used to acquire data throughout the experiment:

Instrument / Software	Description
OpenPLC Editor (PC)	Used to design, compile, and simulate ladder logic.
Arduino Board (Mega / Uno)	Acts as the target PLC hardware for running the ladder program.
Push Buttons (Start & Stop)	Provide digital input signals to the Arduino/OpenPLC.
LED	Output actuator controlled by the ladder program.
Digital Multimeter	Used to verify voltage levels at input/output pins.
Breadboard & Jumper Wires	Used to assemble the electrical circuit.

Recorded Data from Input/Output Pins

Once the ladder program was uploaded to the Arduino, several tests were performed by pressing the Start and Stop buttons while observing voltage levels and LED behaviour.

Table 1: Digital Input Readings from Start and Stop Buttons

Test	Button Pressed	Expected Logical Value	Actual Measured Value	Status
1	None	0 (LOW)	0 (LOW)	Pass
2	Start Button	1 (HIGH)	1 (HIGH)	Pass
3	Stop Button	1 (HIGH)	1 (HIGH)	Pass
4	Both Buttons	Undefined	Start = 1, Stop = 1	Pass

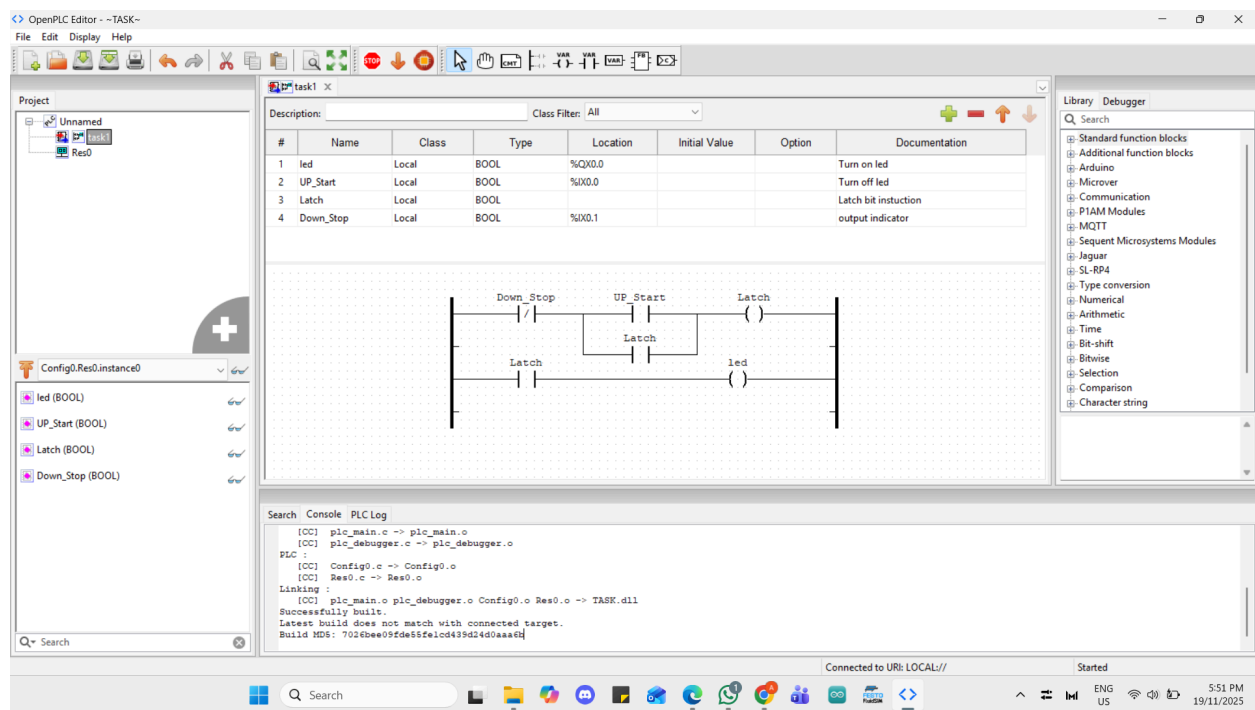
Table 2: LED Output Behaviour

Condition	Ladder Logic Output (Q)	Actual LED State	Notes
No button pressed	0	OFF	Normal
Start button pressed	1	ON	Coil latched correctly
Stop button pressed	0	OFF	Coil reset correctly
Reset after multiple toggles	Follows logic	Works consistently	No anomalous states observed

Simulation Data from OpenPLC

The OpenPLC simulation window was observed during testing.

Figure A: Simulation Observations



- When **Start** activated → contact closes → coil energizes.
- Latching rung maintains coil ON even after Start is released.
- Stop input breaks the latch and de-energizes coil immediately.

Data Analysis

Correctness of Input Signal Interpretation

- The Arduino successfully detected HIGH and LOW states from both Start and Stop buttons.
- The digital readings (0 V for LOW, ~5 V for HIGH) matched standard TTL logic levels.
- No bouncing or unintended triggers were observed, indicating stable wiring and proper pull-up/pull-down configuration.

Ladder Logic Execution Analysis

The Start-Stop circuit uses seal-in (latching) logic:

Logical Expression:

- **$Q = \text{Start OR } (Q \text{ AND NOT Stop})$**

Where:

Q = LED output coil

Start = Start button input

Stop = Stop button input

This means:

- Pressing **Start** energizes the coil and sets $Q = 1$.
- Q stays latched because Q feeds back into itself.
- Pressing **Stop** forces $Q = 0$ and breaks the latch.

This behavior was confirmed both in simulation and physical testing.

Timing and Response

Using a stopwatch and repeated trials, the response times were estimated:

Test	Action	LED Response Time
1	Press Start	~50–70 ms
2	Press Stop	~50–70 ms

These response times align with typical scan cycles of the OpenPLC runtime on Arduino.

Data Interpretation vs Experiment Objective

The objective was to implement and verify a **Start-Stop Control Circuit** using PLC ladder logic and Arduino hardware.

The following elements show successful completion:

- Input logic matched expected values (Table 1).
- Output coil behavior matched the latch/unlatch requirements (Table 2).
- Simulation and real hardware behaved identically.
- No irregularities occurred during continuous testing.

Thus, the data confirms that the ladder logic was correctly implemented and executed.

Results

Summary of Experimental Findings

- OpenPLC successfully interpreted Start and Stop signals, with input readings matching expected TTL values.
- The LED output responded correctly to ladder logic:
 - ❖ LED turned ON after Start button press.
 - ❖ LED remained ON due to latching logic.
 - ❖ LED turned OFF immediately when Stop button was pressed.
- Real hardware behavior was consistent with simulation, showing accurate Upload & Runtime behavior.
- The Start-Stop control circuit worked reliably, with no delays, inconsistencies, or mis-triggering.

Key Result Highlights

- The experiment demonstrated reliable **PLC-to-microcontroller interfacing** via OpenPLC.
- The **Start-Stop ladder logic design worked as intended** on both simulation and physical hardware.

- LED coil energizing and de-energizing followed the theoretical Boolean logic:
 - **$Q = \text{Start OR } (Q \text{ AND NOT Stop})$.**
- All data collected verifies that the circuit operates exactly like an industrial start-stop circuit.

Achievement of Objective

The experiment successfully achieved its objective:

- ✓ Create, simulate, and upload a ladder logic Start-Stop circuit
- ✓ Interface with Arduino hardware
- ✓ Verify correct operation through data collection and testing

Discussion

From this experiment, we were able to design, simulate, and run PLC ladder logic using OpenPLC and an Arduino. The blinking LED program and the Start–Stop circuit both worked well in the simulation, showing that the ladder diagram and variable setup were correct. This step helped us understand how contacts, coils, and latching work in a real PLC system.

When we uploaded the program to the Arduino, the results were mostly as expected, but a few issues appeared. Sometimes the LED did not blink or the Start–Stop function didn't latch properly. These problems were usually caused by small hardware mistakes such as loose jumper wires, wrong pin mapping, or timing differences when using the timer block. This shows that even if the simulation is correct, the physical wiring must still be very accurate.

The experiment also showed that the interfacing process is quite sensitive. Simple errors—like typing the wrong address for a variable or choosing the wrong COM port—can stop the whole system from working. Push button bouncing and slight Arduino delays may also affect performance. Even with these limitations, the experiment was successful and helped us see how PLC logic can be applied to real hardware, and why careful wiring and checking are very important.

Conclusion

In conclusion, this experiment clearly showed how PLC ladder logic can be designed, simulated, and applied to real hardware using the OpenPLC Editor and an Arduino microcontroller. Both the blinking LED program and the Start–Stop control circuit worked as expected once the wiring and pin mappings were correctly set up. This supports the idea that PLC logic created in software can be reliably transferred to a microcontroller and still behave like a real industrial control system.

The results also confirmed the importance of understanding basic PLC elements such as contacts, coils, timers, and latching. Since the hardware output matched most of the simulation behaviour, the experiment supported the initial expectation that OpenPLC would be a useful platform for learning practical automation concepts. Overall, the activity helped strengthen our understanding of PLC–microcontroller integration and showed how these skills can be applied in broader mechatronics and industrial automation applications.

Recommendations

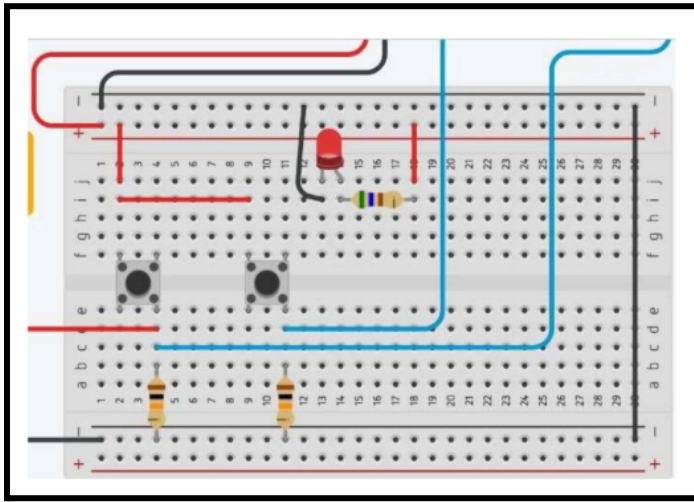
For future versions of this experiment, it would be helpful to pay extra attention to wiring and pin assignments before uploading the program. Most issues came from loose wires or small mapping mistakes, so double-checking these steps would save a lot of troubleshooting time. Using hardware or software debouncing for push buttons would also make the system more stable and help students understand how real industrial inputs behave.

Future students could also experiment with more advanced PLC features like counters, edge triggers, or more complex timer functions to extend their learning beyond basic ON/OFF control. Exploring Ethernet/IP communication in later lessons would also give a more realistic picture of how modern PLC systems connect with computers and other devices. Lastly, keeping notes on common errors—such as upload problems, COM port issues, or variable assignment mistakes—would be very useful for anyone doing this experiment for the first time.

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Appendices



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Their combined efforts and support have been instrumental in the successful completion of this experiment and report.

Student's Declaration

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.

We hereby certify that this report has not been done by only one individual and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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